GSE181919 HNSCC

scRNA-seq Analysis Pipeline

Methods, Statistical Definitions & Rationale

Dataset: GSE181919 — NCBI GEO
Disease: Head & Neck Squamous Cell Carcinoma (HNSCC)
Cells: 51,849 post-QC | 23 patients | 37 samples
Disease stages: NL (normal) → LP (leukoplakia) → CA (cancer) → LN (lymph node)
Research questions:
(1) How does the immune TME change across disease progression?
(2) How does Tcell exhaustion differ between HPV+ and HPV− tumors?

Background: Single-Cell RNA Sequencing (scRNA-seq)

What is scRNA-seq? Single-cell RNA sequencing (scRNA-seq) measures the gene expression profile of thousands of individual cells simultaneously. Unlike bulk RNA-seq which averages expression across all cells in a sample, scRNA-seq reveals cell-type-specific transcriptional programs, rare cell populations, and continuous developmental trajectories.

Why is it used for cancer TME analysis? The tumour microenvironment (TME) is a complex mixture of cancer cells, immune cells, stromal cells, and endothelial cells — all with distinct gene expression programs. scRNA-seq allows us to deconvolute this mixture and study each cell type independently, including how they communicate and evolve during disease progression.

Key concepts in this pipeline:

- **Gene expression matrix:** Rows = genes, Columns = cells, Values = UMI (Unique Molecular Identifier) counts per gene per cell
- **UMI counts:** Molecular barcodes that correct for PCR amplification bias — more accurate than raw read counts
- **UMAP (Uniform Manifold Approximation and Projection):** A dimensionality reduction method that projects high-dimensional gene expression data into 2D for visualisation. Cells with similar transcriptional profiles cluster together
- **Clustering:** Grouping cells with similar gene expression profiles using graph-based algorithms — reveals cell types and subtypes

01 Data Acquisition

Manual download from NCBI GEO database

Purpose & Rationale

GSE181919 was selected because it is one of the largest publicly available HNSCC single-cell datasets, covering four disease progression stages (normal tissue, pre-malignant leukoplakia, primary tumour, and metastatic lymph node) from 23 patients. This longitudinal design allows us to track TME changes across the full disease spectrum — from normal tissue to metastasis — which is essential for answering both research questions.

What was downloaded

File	Contents
GSE181919_UMI_counts.txt.gz	Gene × cell UMI count matrix: 20,000 genes × 54,239 cells
GSE181919_Barcode_metadata.txt.gz	Cell-level metadata: patient.id, sample.id, tissue.type, hpv, cell.type, Age, Gender, subsite

📌 **Gene Expression Matrix**

A numerical table where each row is a gene, each column is a single cell, and each value is the number of UMI counts detected for that gene in that cell. A count of 0 means the gene was not detected — this is called 'dropout' and is a fundamental challenge in scRNA-seq analysis. The high sparsity (>90% zeros) drives many of the analytical choices made in subsequent steps.

02 Seurat Object Creation

01_GSE181919_Creating_Seurat_Object.Rmd

Purpose & Rationale

Seurat is the primary R framework for scRNA-seq analysis. A 'Seurat object' is a standardised data container that stores the count matrix, metadata, dimensionality reductions, and analysis results in a single structured object. Creating this object first establishes the foundation for all downstream analyses.

Key Processing Steps

- **Barcode format fix:** Raw barcodes used '.' (dot) separators; Seurat requires '-' (hyphen). Mismatched formats would prevent metadata from being correctly linked to cells.
- **Protein-coding gene filter:** Filtered from 20,000 to 15,086 genes by removing non-coding RNAs, pseudogenes, and mitochondrial/ribosomal genes. Non-coding genes add noise without contributing to cell type identification.
- **min.cells = 3:** Genes detected in fewer than 3 cells are removed. These are likely technical noise or artefacts rather than biologically meaningful signal.
- **min.features = 200:** Cells with fewer than 200 detected genes are removed. These are likely empty droplets (no cell captured) rather than true single cells.

📌 **Seurat Object**

A specialised R data structure (S4 class) that integrates the raw count matrix with metadata, normalised expression, dimensionality reductions (PCA, UMAP), cluster assignments, and analysis results. Using a standardised container ensures reproducibility and compatibility with Seurat's extensive analysis toolkit. Version 5 (used here) stores assay data in 'layers' rather than single matrices, improving memory efficiency for large datasets.

03 Quality Control & Batch Correction

02_GSE181919_QC_BatchCorrection.Rmd

Purpose & Rationale

Quality control removes technical artefacts (dead cells, doublets, empty droplets) that would confound biological interpretation. Batch correction removes patient-specific technical variation so that the clustering reflects cell biology rather than which patient the cells came from.

QC Filters Applied

Filter	Rationale
MT% > 15% removed (43 cells)	High mitochondrial gene content indicates dying or lysed cells. When a cell lyses, cytoplasmic mRNA leaks out but mitochondrial mRNA (enclosed in mitochondria) is retained, artificially elevating MT%.
Mahalanobis outlier filter (2,344 cells)	Multivariate statistical method that identifies cells whose combination of nCount_RNA and nFeature_RNA are extreme outliers relative to the population distribution. More sensitive than simple min/max thresholds.
Ribosomal filter: SKIPPED	Standard threshold (>10% ribosomal) would remove >97% of cells — HNSCC tumour cells naturally have high ribosomal activity due to active protein synthesis. Applying this filter would introduce severe selection bias.
Gene count: no cells removed	Neither the min (200) nor max (8,000) gene thresholds removed cells, suggesting the authors had already performed basic QC before depositing to GEO.

📌 **Mahalanobis Distance**

A multivariate distance metric that measures how many standard deviations a data point is from the centroid of a distribution, accounting for correlations between variables. Unlike Euclidean distance, it is scale-invariant. In QC, it identifies cells with unusual combinations of total counts and feature numbers — cells that are outliers in multivariate space even if they appear normal on individual metrics alone. The worst 5% of cells by Mahalanobis distance are removed.

Normalisation

LogNormalize (scale factor = 10,000): Raw UMI counts are divided by the total counts per cell, multiplied by 10,000 (making counts 'per 10,000 UMIs'), then log-transformed (log1p). This corrects for differences in sequencing depth between cells (some cells are sequenced more deeply than others) and compresses the expression scale to reduce the influence of highly expressed genes.

📌 **LogNormalization Formula**

Normalised value = log(1 + (raw_count / total_cell_counts × 10,000)). The +1 inside the log prevents log(0) errors for zero-count genes. This is the most widely used normalisation method for scRNA-seq because it is simple, interpretable, and effective for downstream analyses including clustering and visualisation.

Dimensionality Reduction

- **HVG selection (2,500 genes):** Highly Variable Genes (HVGs) are genes that show the most variation across cells — they carry the most information about cell identity. Using only HVGs for PCA reduces noise from constitutively expressed housekeeping genes.
- **PCA — 100 components, 50 selected:** Principal Component Analysis decomposes the expression matrix into orthogonal components ranked by variance explained. The first 50 PCs collectively explain 89.3% of variance. Using 50 PCs balances information capture with noise reduction.
- **UMAP:** Applied to the 50 PC embeddings to generate a 2D visualisation. UMAP preserves both local (neighbourhood) and global (cluster) structure better than t-SNE for large datasets.

> 📌 **Principal Component Analysis (PCA)**
>
> A linear dimensionality reduction technique that identifies orthogonal axes (principal components) that capture the maximum variance in the data. Each PC is a linear combination of gene expression values. PC1 captures the most variance, PC2 the second most, etc. In scRNA-seq, PCA compresses 15,086 gene dimensions into 50-100 meaningful dimensions, removing technical noise while retaining biological signal. The elbow plot of variance-per-PC guides the selection of the optimal number of PCs.

Batch Correction with Harmony

Harmony is an iterative algorithm that corrects PCA embeddings for batch effects by moving cells in PCA space to reduce their correlation with batch variables (here: patient.id). It operates in reduced-dimensional space (PCA) rather than on raw counts, making it computationally efficient and biologically conservative.

> 📌 **Harmony Batch Correction**
>
> Harmony works by iteratively: (1) performing soft clustering of cells, (2) computing per-cluster, per-batch correction factors, and (3) applying these factors to shift cell positions in PCA space. It does not modify gene expression values — only the embeddings used for UMAP and clustering. Same-batch fraction (the proportion of nearest neighbours from the same patient) dropped from 0.470 to 0.247 after correction, a 47.4% reduction confirming effective batch removal.

04 CellTypist Automated Annotation

03_GSE181919_CellTypist_Annotation.Rmd

Purpose & Rationale

Cell type annotation assigns a biological identity to each cluster. Manual annotation requires expert knowledge and is subjective. CellTypist provides automated, reproducible annotation using a pre-trained logistic regression classifier trained on curated reference datasets from human tissues.

Why CellTypist over manual annotation?

- **Reproducibility:** Automated methods produce the same result every time; manual annotation varies between researchers
- **Scale:** 51,849 cells across 29 clusters would require extensive manual curation of marker genes
- **Reference quality:** The Immune_All_High.pkl model is trained on 20 human tissues with expert-validated labels
- **Validation:** CellTypist predictions were compared against author-provided labels to quantify agreement

> 📌 **Logistic Regression Classifier (CellTypist)**
>
> CellTypist uses a multi-class logistic regression model trained on reference single-cell data. For each cell, it computes the probability of belonging to each reference cell type based on normalised gene expression. The highest-probability label is assigned. The model was trained on the Immune_All_High reference (spanning 20 human tissues, covering immune and stromal populations) which directly covers 8 of 10 GSE181919 cell types. Malignant cells are predicted as Epithelial (biologically correct — HNSCC malignant cells are of epithelial origin; the model has no 'malignant' category).

Technical Note: Why Python via reticulate?

CellTypist is a Python package. Running 51,849 cells × 15,086 genes through the Python standalone crashed on 16 GB RAM during the scaling step. By calling Python from R via the reticulate package, memory is shared across both environments, avoiding the crash. Per-cell prediction (majority_voting = FALSE) was used because majority voting requires a neighbour graph computation that also exceeded RAM limits.

Validation Results

Cell Type	CellTypist Agreement with Author Labels
Mast cells	98.1%
pDCs (plasmacytoid Dendritic Cells)	97.9%
Plasma cells	95.7%
Fibroblasts	94.6%
T cells	94.3%
Macrophages	92.4%
B cells	90.3%
Endothelial cells	89.6%
Malignant cells	83.3% (predicted as Epithelial — biologically expected)

05 TME Composition Analysis

04_GSE181919_Tumor_Microenvironment_Analysis.Rmd

Purpose & Rationale

Characterising the proportional composition of the TME across disease stages is the foundational descriptive analysis. It reveals how the balance of cell types shifts from normal tissue to metastasis, and how HPV status shapes the immune landscape — directly addressing Research Question 1.

Analyses Performed

- **Cell type proportions by tissue stage (NL/LP/CA/LN):** Stacked and grouped bar plots showing the percentage of each cell type at each stage. Reveals the stromal-to-immune shift: Fibroblasts 52%→2%, T cells 25%→57% across NL→LN.
- **TME compartment ratios:** Cells grouped into three biological compartments: Immune (T cells, B/Plasma, Macrophages, DCs, Mast cells), Stromal (Fibroblasts, Endothelial, Myocytes, Epithelial), Malignant. The immune compartment expands from 33% to 95% across NL→LN.
- **HPV+ vs HPV− comparison (CA only):** HPV+ tumours are T cell-rich (50% vs 27%) while HPV− tumours are Macrophage-rich (21% vs 5%), reflecting fundamentally different immune programs driven by viral antigen presence versus absence.
- **Canonical marker gene dot plot:** 40 marker genes across 10 cell types validates annotation quality. Dot size = fraction of cells expressing the gene; dot colour = average expression level.

> 📌 **Dot Plot for Marker Gene Validation**
>
> A dot plot encodes two dimensions of information simultaneously: (1) the percentage of cells expressing a gene (dot size) and (2) the average normalised expression level (dot colour). A good cell type marker shows a large, dark dot in its corresponding cell type and small/pale dots in all others. This provides quick visual validation that the cell type annotations are biologically correct and that canonical markers are specifically expressed where expected.

06 T Cell Exhaustion Analysis

05_GSE181919_T_Cell_Exhaustion_Analysis.Rmd

Purpose & Rationale

T cell exhaustion is a dysfunctional state characterised by progressive loss of effector function and upregulation of inhibitory receptors. It is the primary mechanism by which tumours evade immune destruction. This step quantifies exhaustion states at single-cell resolution and compares them between HPV+ and HPV− tumours — directly addressing Research Question 2.

T Cell Subclustering

Why re-cluster T cells separately? Clustering all 51,849 cells together produces clusters dominated by the most abundant cell types. Subsetting T cells (17,533 cells) and re-clustering at higher resolution (0.5) reveals fine-grained T cell subtypes — CD8+ cytotoxic, CD4+ helper, Treg, NKT-like, proliferating (MKI67+) — invisible in the full-dataset clustering.

UCell Scoring: Quantifying Functional States

UCell assigns a continuous score to each cell representing its membership in a functional gene signature. Unlike discrete cluster labels, scores allow for gradient analysis — cells can be partially exhausted, transitioning between states.

> 📌 **UCell (Rank-Based Gene Set Scoring)**
>
> UCell computes gene set scores using the Mann-Whitney U statistic applied to per-cell gene rankings. For each cell, all genes are ranked by expression. The score for a gene set is the fraction of set genes that rank higher than expected by chance, adjusted for set size. Key advantages over other scoring methods: (1) rank-based — robust to outliers and differences in total counts; (2) no negative control gene set required; (3) scores are comparable across cells with different sequencing depths. Scores range from 0 (no signature activity) to 1 (complete signature expression).

Gene Signatures Used

State	Signature Genes
Exhausted	PDCD1 (PD-1), HAVCR2 (TIM-3), LAG3, TIGIT, CTLA4
Effector	GZMB (Granzyme B), PRF1 (Perforin), IFNG, TNF, GZMA
Naive / Memory	TCF7 (TCF1), CCR7, SELL (CD62L), LEF1, IL7R
Treg (Regulatory T cell)	FOXP3, IL2RA (CD25), IKZF2, CTLA4, TNFRSF18
Progenitor_Tex (Stem-like exhausted)	TCF7, PDCD1, SLAMF6, CXCR5, TOX

Statistical Testing: Wilcoxon Rank-Sum Test

HPV+ vs HPV− comparisons of UCell scores used the Wilcoxon rank-sum test (also called Mann-Whitney U test).

📌 **Wilcoxon Rank-Sum Test**

A non-parametric test that compares the distributions of two groups without assuming normality. It tests whether values from one group tend to be larger than the other. Null hypothesis: the two distributions are identical. Applied here to compare UCell score distributions between HPV+ and HPV− T cells. Chosen over t-test because UCell scores are not normally distributed (bounded 0-1, often skewed). All 5 state comparisons were significant at $p < 0.0001$.

07 Differential Expression Analysis (DEG)

06_GSE181919_Differential_Expression_Analysis.Rmd

Purpose & Rationale

Differential expression analysis identifies which genes are significantly upregulated or downregulated between conditions (e.g., tumour vs normal, HPV+ vs HPV−) within each cell type. This reveals the molecular mechanisms driving phenotypic differences and identifies candidate biomarkers and therapeutic targets.

Part 1: Cell Type Marker Genes (FindAllMarkers)

Method: Seurat's FindAllMarkers uses the Wilcoxon rank-sum test to compare each cluster against all other clusters. Returns genes with significant positive fold change (logFC > 0.25, adjusted $p < 0.05$, min.pct = 0.1). Identifies 9,206 marker genes across 10 cell types.

Part 2: Pseudobulk DESeq2 (Primary DE Method)

For condition comparisons (NL vs CA, CA vs LN, HPV+ vs HPV−), pseudobulk DESeq2 was used instead of single-cell tests. This is the methodologically superior approach for multi-patient single-cell data.

📌 **Pseudobulk Analysis**

Pseudobulk analysis aggregates (sums) raw counts from all cells of a given cell type within each patient sample, creating one 'bulk-like' expression profile per patient per cell type. These patient-level aggregates are then analysed with DESeq2 as if they were bulk RNA-seq samples, treating patients as biological replicates. This is statistically correct because individual cells from the same patient are not independent — they share the same genome, epigenome, and microenvironment. Treating cells as independent samples (as naive single-cell DE methods do) massively inflates sample size and produces false positives.

📌 **DESeq2 (Negative Binomial Regression)**

DESeq2 models raw count data using a negative binomial distribution, which handles the overdispersion (higher variance than expected for a Poisson distribution) typical of RNA-seq count data. It uses 'shrinkage estimation' to stabilise fold-change estimates for lowly expressed genes. For each gene it tests H_0: log2FoldChange = 0. Multiple testing correction uses the Benjamini-Hochberg procedure to control the False Discovery Rate (FDR). Significant genes: $|\log2FC| > 0.5$, adjusted p-value (padj) < 0.05.

Key Findings

- **NL vs CA:** Fibroblasts most reprogrammed (5,388 DEGs) — consistent with their 52%→9% proportional collapse. CXCL13 is the top upregulated gene in both T cells AND Fibroblasts, revealing convergent TLS (tertiary lymphoid structure) formation.
- **CA vs LN:** Only 30 T cell DEGs — all upregulated in LN. The immune program established in the primary tumour is fully maintained through metastasis, confirming the tumour 'educates' the lymph node.
- **HPV+ vs HPV−:** HPV+ upregulates heat shock proteins (HSPA1A, DNAJB1) — viral protein folding stress. HPV− upregulates chemokines (CXCL3, CXCL8, CXCL2) — broader inflammatory program without antigen specificity.

📌 **Volcano Plot**

A scatter plot used to visualise differential expression results. The x-axis shows log2 fold change (positive = upregulated in condition A; negative = upregulated in condition B). The y-axis shows -log10(adjusted p-value) — higher points are more statistically significant. Genes in the upper-right quadrant are significantly upregulated; upper-left are significantly downregulated. The 'volcano' shape emerges because highly significant genes tend to also have large fold changes.

08 CellChat — Cell-Cell Communication Analysis

07_GSE181919_CellChat_Cell-Cell_Communication_Analysis.Rmd

Purpose & Rationale

Gene expression tells us what each cell is doing. CellChat tells us how cells are talking to each other. By modelling ligand-receptor (L-R) interactions between cell types, we can identify which cells send signals, which receive them, and which signalling pathways drive TME organisation, immune evasion, and T cell exhaustion.

What is CellChat?

CellChat infers, analyses, and visualises intercellular communication from single-cell data. For each L-R pair in its curated database, it computes a communication probability between every pair of cell types based on the expression levels of the ligand in sender cells and the receptor in receiver cells.

> 📌 **CellChat Communication Probability**
>
> For each L-R pair, CellChat computes: P(L→R) = f(expression_ligand) × f(expression_receptor) × S, where f() is a non-linear transformation of trimean gene expression and S is a scaling factor. The trimean (average of median + two quartiles) is more robust than mean expression and reduces the influence of outlier-expressing cells. Probabilities are computed separately for each cell type pair, creating a communication network. Pathways are groups of related L-R pairs (e.g., 'COLLAGEN' pathway = all collagen-integrin pairs).

Database Coverage

Database	Coverage
Secreted Signaling	Cytokines, chemokines, growth factors secreted into the extracellular space
ECM-Receptor	Extracellular matrix proteins (collagens, laminins, fibronectin) binding cell surface receptors
Cell-Cell Contact	Direct membrane-to-membrane interactions (e.g., MHC-peptide-TCR, PD-L1-PD-1)

Key Analyses

- **Circle plots:** Visualise the signalling network between cell types. Arrow thickness = interaction strength; node size = total interaction weight; self-loops = autocrine signalling
- **Centrality scatter plots:** Each cell type plotted by outgoing (sender) vs incoming (receiver) communication strength. Reveals dominant senders, receivers, and mediators
- **Bubble plots:** Show which specific L-R pairs are active in each signalling pathway. X-axis = cell type pairs; Y-axis = L-R pairs; bubble size = communication probability
- **Differential heatmaps:** Show which interactions are stronger in condition A vs condition B. Red = stronger in A; Blue = stronger in B

Technical Fix: subsetCellChat()

NL tissue lacks Malignant cells; CA tissue lacks Epithelial cells and Myocytes. When merging CellChat objects for comparison, non-conformable arrays errors occur if the cell types differ between objects. Solution: subsetCellChat() on each object to retain only cell types present in both conditions before calling mergeCellChat().

Key Mechanistic Findings

Circuit	Mechanism
CAF Broadcasting	Fibroblasts → ALL cell types via COLLAGEN/LAMININ/FN1/TGFb/VEGF. COL1A1/1A2/4A1/4A2 → integrin receptors on Malignant cells at highest intensity — invasion scaffold formation
TAM Integration	All cell types → Macrophages via MIF, GALECTIN, CXCL, PLAU — pro-tumour polarisation. Macrophages integrate diverse immunosuppressive signals
T Cell Exhaustion	All cell types → T cells via MHC-I (ubiquitous antigen presentation). LGALS9 → HAVCR2 (Galectin-9 → TIM-3) from DCs = primary checkpoint axis
HPV Immune Evasion	HPV+ tumours suppress MHC-II (HLA-DR/DQ/DP → CD4) from DCs — evades CD4+ helper T cell surveillance. Absent in HPV− tumours

09 Pseudotime Trajectory Analysis (Monocle3)

08_GSE181919_Trajectory_Analysis_Monocle3.Rmd

Purpose & Rationale

Trajectory analysis orders cells along a continuous developmental path — called pseudotime — based on their transcriptional similarity. Unlike discrete cluster labels, pseudotime reveals the gradual gene expression changes that occur as cells transition from one state to another. This is critical for studying T cell exhaustion, which is a continuous process (not a binary switch) from naïve → progenitor exhausted → terminally exhausted states.

📌 **Pseudotime**

Pseudotime is a computational measure of a cell's position along a developmental or differentiation trajectory. It is not real time — it represents transcriptional distance from a defined root state. A cell with pseudotime = 0 is at the root (most naïve/undifferentiated), while cells with high pseudotime are at the end of the trajectory (most differentiated/exhausted). Pseudotime is computed by projecting cells onto a principal graph and measuring their geodesic distance from the root.

Monocle3 Pipeline

Step	Purpose
as.cell_data_set()	Convert Seurat object to Monocle3 CDS (cell_data_set) format
preprocess_cds()	PCA within Monocle3 (required for Seurat v5 compatibility — Seurat v5 PCA format is incompatible with Monocle3 graph_test)
reduce_dimension()	UMAP within Monocle3 (separate from Seurat UMAP — uses Monocle3-internal embeddings)
cluster_cells()	Leiden-based clustering within Monocle3
learn_graph()	Learns the principal graph (trajectory structure) using SimplePPT algorithm
order_cells()	Assigns pseudotime values by setting root cells (Naive_Memory T cells) and computing geodesic distances

📌 **Principal Graph Learning (SimplePPT)**

Monocle3 learns the trajectory topology using a Reversed Graph Embedding approach. It fits a principal tree (a connected acyclic graph) to the UMAP embedding by iteratively: (1) projecting cells onto the graph, (2) updating graph node positions to minimise the distance between cells and their projections. The result is a principal graph whose branches represent diverging cell fates. Parameters: ncenter (number of graph nodes) and minimal_branch_len (minimum branch length) control graph resolution. Pseudotime = geodesic distance along the graph from the root node.

Moran's I Graph Autocorrelation Test

Purpose: Identifies genes whose expression is significantly structured along the trajectory — i.e., not randomly distributed but showing spatial autocorrelation in pseudotime space. These are the trajectory-variable genes**.**

> 📌 **Moran's I Statistic**
>
> Originally a spatial statistics measure, Moran's I quantifies the degree of spatial autocorrelation — whether nearby values are more similar than expected by chance. Adapted for single-cell trajectories, it measures whether a gene's expression along the principal graph is more structured than random. Values range from -1 (perfect negative autocorrelation) to +1 (perfect positive autocorrelation). A high Moran's I (near +1) means cells that are close in pseudotime have similar expression — the gene is trajectory-variable. Statistical significance is assessed by permutation testing; q-value (BH-corrected p-value) < 0.05 indicates a significant trajectory gene. 10,833 genes passed this threshold in the T cell analysis.

Gene Module Analysis

find_gene_modules() groups trajectory-variable genes into co-expression modules using Louvain community detection on a gene-gene similarity graph. Each module represents a gene programme that is co-activated at the same point along the pseudotime trajectory.

> 📌 **Louvain Community Detection**
>
> A graph-based clustering algorithm that partitions nodes into communities by maximising modularity — a measure of how much the observed connections between nodes within a community exceed what is expected by chance. Applied to gene-gene similarity graphs (where edge weight = correlation of expression along pseudotime), it identifies groups of co-expressed genes. The 'resolution' parameter controls community granularity: lower resolution → fewer, larger modules; higher resolution → more, smaller modules. Resolution = 0.1 was used here to obtain biologically meaningful separation (6 modules vs 3 at lower resolution).

6 Gene Modules Identified

Module	Identity & Key Genes
Module 1	Effector/Exhaustion — CXCL13, LAG3, HAVCR2, GZMB, PRF1, TIGIT, PDCD1, TOX. NL→LN 3× increase
Module 2	Proliferation — STMN1, TYMS, TUBA1B (cell cycle genes). Peak in Progenitor_Tex — stem pool cycles
Module 3	Intermediate/Stress — TCF7, HSP90AA1, HSPA1A. Naïve-to-exhaustion transition with proteotoxic stress
Module 4	Minor contamination — FXYD3, S100A14. Low expression, likely residual non-T cell signal
Module 5	Treg programme — FOXP3, TNFRSF18, ACTB. Exclusively elevated in Treg state (CA/LN specific)

Module 6	Tumour-associated signal — ELF3, MAGEA4, MDK. Tumour antigen expression in CA-infiltrating T cells

Technical Fixes Required

- **Seurat v5 + Monocle3 v1.4.26 incompatibility:** graph_test() requires Monocle3-native PCA. As.cell_data_set() transfers Seurat's PCA, which is stored in a different matrix format in v5 (dgCMatrix vs dense matrix). Fix: preprocess_cds() → reduce_dimension() replaces the transferred PCA with Monocle3-native PCA before calling graph_test().
- **gene_short_name column:** SeuratWrappers v0.4.0 does not populate rowData()$gene_short_name during as.cell_data_set() conversion. plot_cells(genes=) and find_gene_modules() both require this column. Fix: manually assign rowData(cds)$gene_short_name <- rownames(cds) after conversion.
- **aggregate_gene_expression() format change:** The cell_group_df parameter format changed between monocle3 versions. Fix: manual sapply() aggregation computing mean module expression per cell type group, bypassing the built-in function entirely.

Key Cross-Analysis Findings

Finding 1: The CXCL13 Convergence — TLS Formation

What: CXCL13 is the #1 upregulated gene in both T cells AND Fibroblasts (NL vs CA, Script 07). It peaks early in the T cell exhaustion trajectory at pseudotime ~0.5 (Script 09) and is 50% higher in HPV+ T cells than HPV− (Scripts 06, 09).

Why it matters: CXCL13 (B lymphocyte chemoattractant) recruits B cells and organises Tertiary Lymphoid Structures (TLS). The convergence of exhausted T cells and Cancer-Associated Fibroblasts (CAFs) on the same chemokine signal directly explains why B/Plasma cells expand from 5% (NL) to 18% (CA). TLS are associated with improved responses to immunotherapy — this finding connects T cell exhaustion biology to tumour immunology at a mechanistic level.

Finding 2: Three-Way Exhaustion Confirmation

What: T cell exhaustion is confirmed by three independent methods: (1) UCell scores higher in HPV− (Script 06), (2) CellChat MHC-I/GALECTIN/TIM-3 exhaustion pathways intensify NL→CA (Script 08), (3) Trajectory Module 1 (Exhaustion) increases 3× NL→LN (Script 09).

Why it matters: Independent confirmation by transcriptional scoring, cell-cell communication analysis, and pseudotime trajectory analysis provides high confidence in the exhaustion phenotype. Each method measures a different aspect: UCell = static gene expression, CellChat = intercellular signals driving exhaustion, Trajectory = dynamic progression of exhaustion over pseudotime.

Finding 3: The HPV+ Exhaustion Paradox

What: HPV− T cells score higher on exhaustion (Script 06, $p < 0.0001$) yet HPV+ T cells have 2× more HAVCR2 (TIM-3) and 50% more CXCL13 expression along pseudotime (Script 09). HPV+ tumours have fewer CellChat exhaustion interactions but more organised MHC-II suppression (Script 08).

Why it matters: HPV+ creates a 'deeper but organised' exhaustion — fewer total exhausted cells but those that exhaust do so more completely (higher HAVCR2) and are better organised for TLS formation (higher CXCL13). HPV− creates 'broader but shallower' exhaustion — more cells are partially exhausted but the programme lacks organisation. HPV+ better prognosis despite more exhaustion is explained by the CXCL13-TLS axis: structured antigen presentation in TLS can partially reverse exhaustion.

Finding 4: TIGIT as Primary Therapeutic Target

What: Among all checkpoint markers, TIGIT is the only one that monotonically increases along the entire T cell pseudotime trajectory (Script 09). HAVCR2 and LAG3 show bimodal peaks; PDCD1 is broadly expressed and plateaus early; TCF7 declines. TIGIT is also the highest-intensity signal in the terminal exhaustion cluster.

Why it matters: A monotonically increasing checkpoint represents a continuously progressive therapeutic target — blocking TIGIT at any point along the exhaustion trajectory would intercept cells before they reach the terminal state. PDCD1 (PD-1) and HAVCR2 (TIM-3), already established therapeutic targets, show non-monotonic dynamics suggesting they act at specific exhaustion stages rather than continuously.

Finding 5: Macrophage Full Reprogramming — Dual Confirmation

What: Macrophages are the top signalling receiver in CA (CellChat incoming centrality ~9.5, Script 08) and have the highest median pseudotime among all cell types (~4.5, Script 09).

Why it matters: CellChat shows macrophages receive the most diverse signals (MIF, GALECTIN, CXCL, PLAU, COLLAGEN). Pseudotime confirms they are maximally transcriptionally reprogrammed from the NL T cell root. Together, these findings confirm that tumour-associated macrophages (TAMs) are the most comprehensively rewired cell type in the HNSCC TME — receiving signals from every other cell type and reflecting the accumulated effect of this reprogramming in their transcriptional state.

Statistical Methods Glossary

Dimensionality Reduction

📌 PCA — Principal Component Analysis

Linear transformation that finds orthogonal axes of maximum variance in high-dimensional data. In scRNA-seq, reduces 15,086 gene dimensions to ~50 meaningful components. Each PC is a weighted linear combination of gene expression values. PC1 captures the most variance (largest differences between cell types), subsequent PCs capture progressively less. Computationally efficient; used as input to UMAP and clustering rather than for direct visualisation.

📌 UMAP — Uniform Manifold Approximation and Projection

Non-linear dimensionality reduction that preserves both local (within-cluster) and global (between-cluster) structure. Constructs a high-dimensional fuzzy simplicial complex (topological representation) of the data, then optimises a low-dimensional embedding to have similar topology. Better than t-SNE for large datasets and at preserving global structure. Warning: UMAP distances between clusters are not strictly interpretable — cluster separation in 2D does not directly correspond to transcriptional distance.

Clustering

📌 Leiden / Louvain Community Detection

Graph-based clustering algorithms that operate on a k-nearest-neighbour (kNN) graph of cells. Each cell is connected to its k most similar cells (by PCA distance). The algorithm partitions this graph into communities by maximising modularity. Resolution parameter controls granularity: higher resolution → more, smaller clusters. Leiden is an improved version of Louvain that guarantees well-connected communities. Used in both Seurat (FindClusters) and Monocle3 (cluster_cells).

Normalisation

📌 LogNormalization (NormalizeData)

Corrects for sequencing depth differences between cells. Formula: log(1 + count/total_counts × 10,000). The division by total_counts scales all cells to the same 'sequencing depth'. Multiplication by 10,000 makes values human-readable. log1p transformation compresses the range and makes the distribution more Gaussian-like, which is better for downstream analyses.

Differential Expression

📌 Wilcoxon Rank-Sum Test (non-parametric)

Tests whether values from group A tend to be larger than group B without assuming normality. Converts values to ranks, then tests if the rank sums differ between groups. Used for: (1) FindAllMarkers (single-cell DE), (2) UCell score comparisons between HPV groups. Robust to outliers and skewed distributions common in single-cell data. Multiple testing correction by Bonferroni (FindAllMarkers) or Benjamini-Hochberg (UCell comparisons).

📌 DESeq2 — Negative Binomial Regression

Models count data (integers ≥ 0) using a negative binomial distribution, which handles overdispersion (variance > mean, common in RNA-seq). Uses shrinkage estimation (empirical Bayes) to stabilise effect size estimates for lowly expressed genes. Tests H_0: log2FC = 0 using a Wald test. Pseudobulk aggregation (summing counts per patient per cell type) provides true biological replicates, preventing the pseudo-replication that inflates significance in naive single-cell DE. FDR control by Benjamini-Hochberg. Used for: NL vs CA, CA vs LN, HPV+ vs HPV− comparisons.

Gene Set Scoring

📌 UCell (Rank-Based Module Scoring)

Computes gene set scores using the Mann-Whitney U statistic on per-cell gene rankings. For cell c and gene set G: score = (U − minimum_U) / (maximum_U − minimum_U), where U is computed over the ranks of genes in G vs all other genes. Advantages: rank-based (robust to depth differences and outliers), no reference required, scores are comparable across cells and datasets. Range: 0-1.

Communication Analysis

📌 CellChat Communication Probability

For L-R pair (L, R) between sender cell type S and receiver cell type T: P(S→T) = trimean(L in S) × trimean(R in T) × hill_transform. Trimean = (Q1 + 2×median + Q3)/4 — robust to outliers. Hill transform is a non-linear function that saturates at high expression. Pathways aggregate multiple L-R pairs. Statistical significance: permutation test comparing observed probability to null distribution from 100 cell-label shuffles.

Trajectory Statistics

📌 Moran's I (Spatial Autocorrelation)

Moran's I = $[N/W] \times [\Sigma_i \Sigma_j w_{ij}(x_i - \bar{x})(x_j - \bar{x})] / [\Sigma_i(x_i - \bar{x})^2]$, where N = number of cells, W = sum of weights, w_{ij} = graph connectivity weight between cells i and j, x_i = gene expression in cell i. In the trajectory context, w_{ij} is derived from the principal graph adjacency. High Moran's I → cells close on the trajectory have similar expression → gene varies along pseudotime. Significance by permutation (1,000 shuffles), p-values corrected by Benjamini-Hochberg.

📌 **Louvain Community Detection (Gene Modules)**

Applied to gene-gene correlation matrix (Pearson correlation of expression along pseudotime). Genes are nodes; edges weighted by correlation strength. Louvain maximises Q = Σ_communities [edges_within / total_edges − (degree_sum/2total_edges)²]. Resolution parameter: 0.1 used here. At this resolution, 6 biologically meaningful modules were identified — corresponding to Effector/Exhaustion, Proliferation, Stress/Intermediate, Minor contamination, Treg, and Tumour-associated programmes.

Batch Correction

📌 **Harmony — Iterative Batch Correction in PCA Space**

Harmony operates on PCA embeddings (not raw counts). Algorithm: (1) Initialise cluster centroids using fuzzy K-means. (2) For each batch, compute a correction factor = difference between cluster centroid in this batch vs overall centroid. (3) Apply correction to shift cells. (4) Repeat until convergence. Convergence criterion: change in objective function < tolerance. Key advantage: preserves biological variation while removing technical batch effects. Does not modify gene expression values — only the PCA embeddings used for UMAP and clustering.

Software & Versions

Package	Version	Primary Use
Seurat	5.x	Core scRNA-seq framework — object management, normalisation, clustering, visualisation
Harmony	1.x	PCA-space batch correction by patient
CellTypist	1.x	Automated cell type annotation (Python, via reticulate)
UCell	2.x	Rank-based gene set scoring for functional states
DESeq2	1.x	Pseudobulk differential expression analysis
CellChat	≥2.1.0	Ligand-receptor cell-cell communication inference
monocle3	1.4.26	Pseudotime trajectory analysis
SeuratWrappers	0.4.0	Seurat-to-Monocle3 CDS conversion
ggplot2	3.x	Publication-quality data visualisation
patchwork	1.x	Multi-panel figure composition
pheatmap	1.x	Heatmap generation (modules, pathways)
viridis	0.x	Perceptually uniform colour scales
R	4.5	Statistical computing environment